IAUS406 98S-like ● SNII #054

Complex pre-explosion mass loss in red supergiants, as illuminated by the SN 1998S-like SN 2024cld

Early-time flash ionisation → persistent interaction with disk-like CSM

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FIG 01 ● Deep NOT/ALFOSC *gri* image of SN 2024cld

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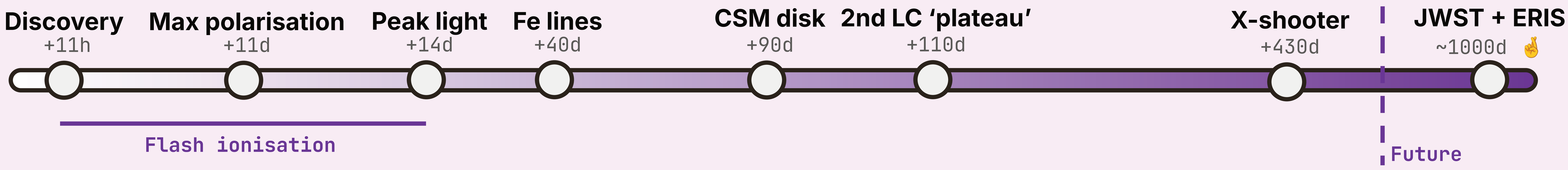
KEY RESULT

SN 2024cld was discovered just **11h post-explosion** via the GOTO-FAST program, likely the earliest of the elusive 98S-like SNe. Continuous monitoring over the next 200d revealed complex, aspherical CSM driving enduring interaction, attributable to a **disk of material shed by binary interaction**, with spectroscopy and polarimetry pinning the geometry. A prodigious dust component at late time further underscores **disk-interacting SNe as potent dust factories** – with dust properties and formation to be probed in SN 2024cld via **awarded JWST Cycle 5 time**.

Continuous coverage: from 11h post=explosion to 200d (and beyond)

FIG 02 ● Timeline



01 Discovery and extended flash-ionisation phase

Discovered by GOTO and classified on the INT just **11 hours post-explosion**, as part of the **GOTO-FAST** transient follow-up program – pairing a tailored high-cadence sky survey, with visitor mode allocations for responsive classifications.

SN 2024cld initially looked like an ordinary flash-ionised (FI) SN II: but with a long-duration FI phase, extended rise to peak, and continued narrow emission even after two weeks, interaction likely dominates the evolution.

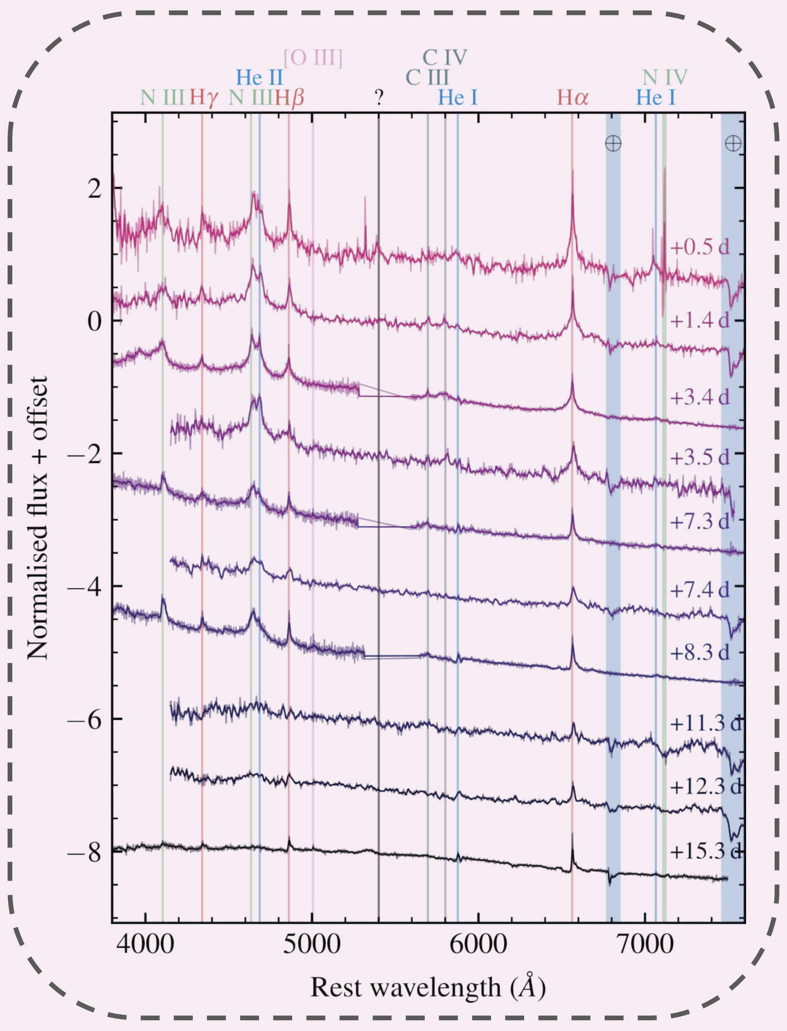


FIG 03 ● *Early-time flash features of SN 2024cld, with He II, C III/IV, and NIII*

03 Light curve evolution

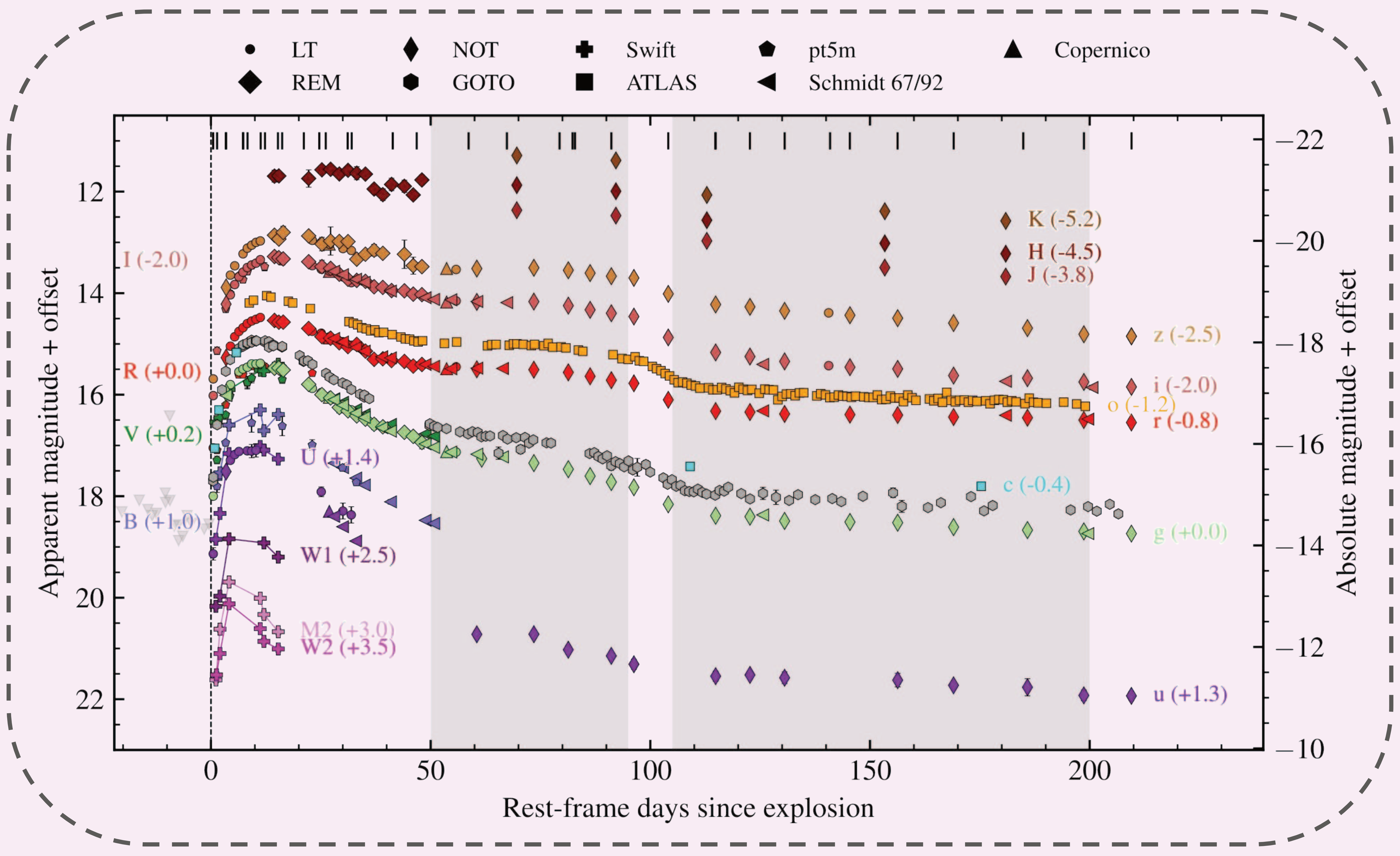


FIG 07 ● *Light curve for SN 2024cld. The SN reaches a peak absolute magnitude of* ***g=-17.58*** *at* ***14.6 days*** *post-explosion. The late-time evolution is entirely driven by CSM interaction*

02 An emerging multi-component CSM

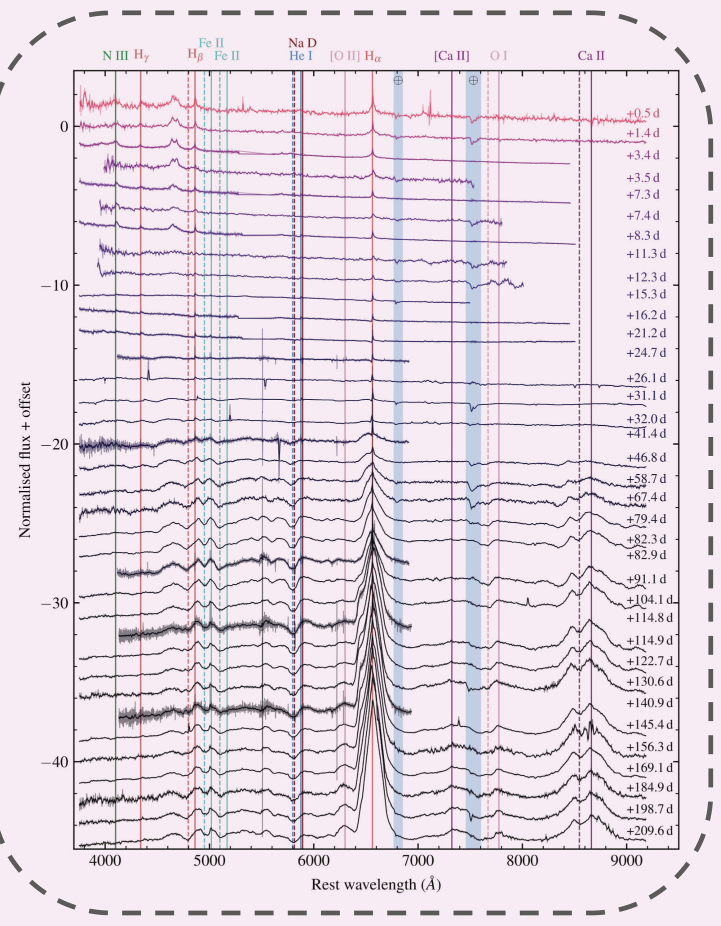


FIG 04 ● *Full spectral series of SN 2024cld, from 11h post-explosion to +200d*

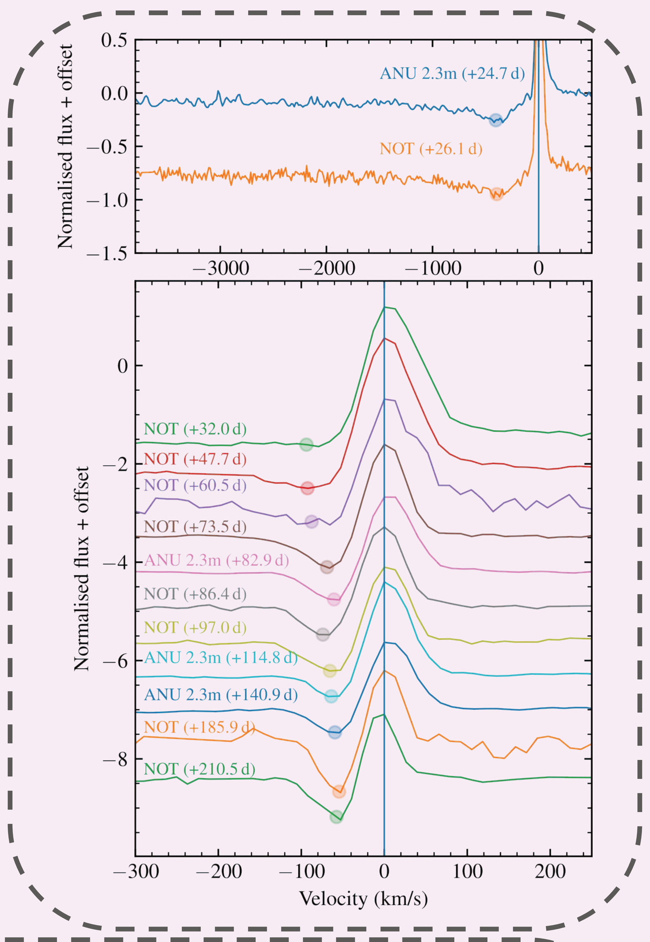


FIG 05 ● *High-resolution spectroscopy reveals a narrow P Cygni profile, associated with pre-shock wind*

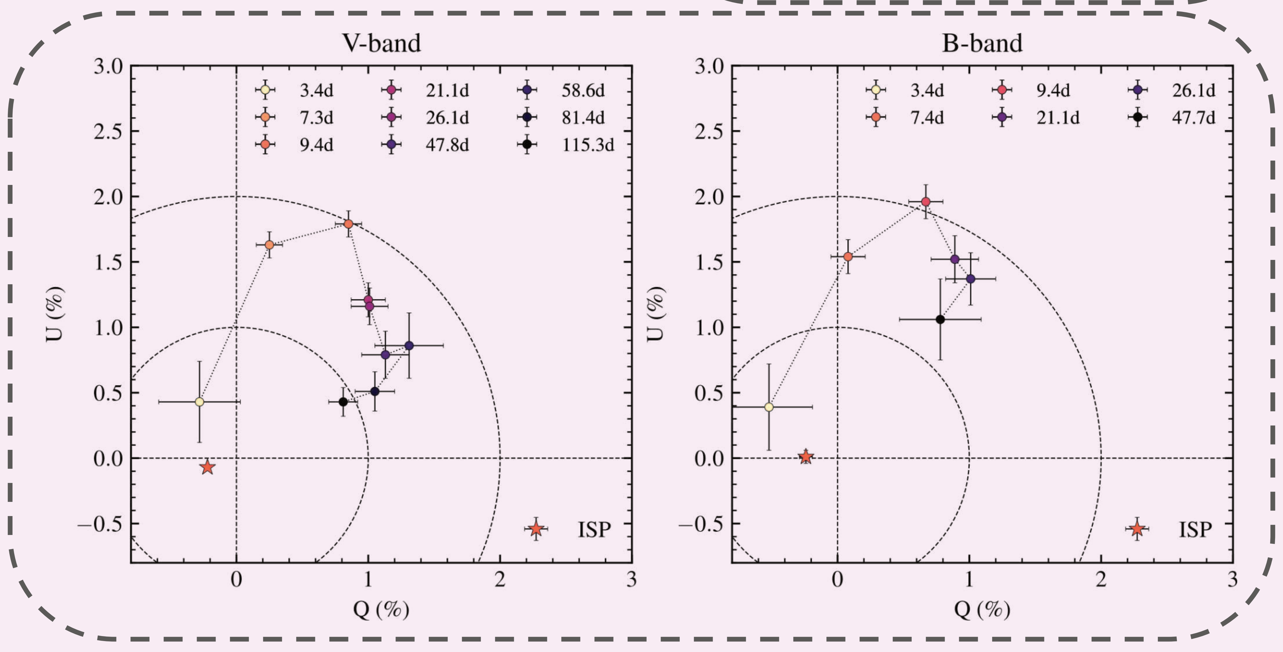


FIG 06 ● *Polarisation evolution of SN 2024cld, showing a rise to a peak of 2% around 10 days post-explosion, followed by a slow decline and rotation.*

05 Scenario

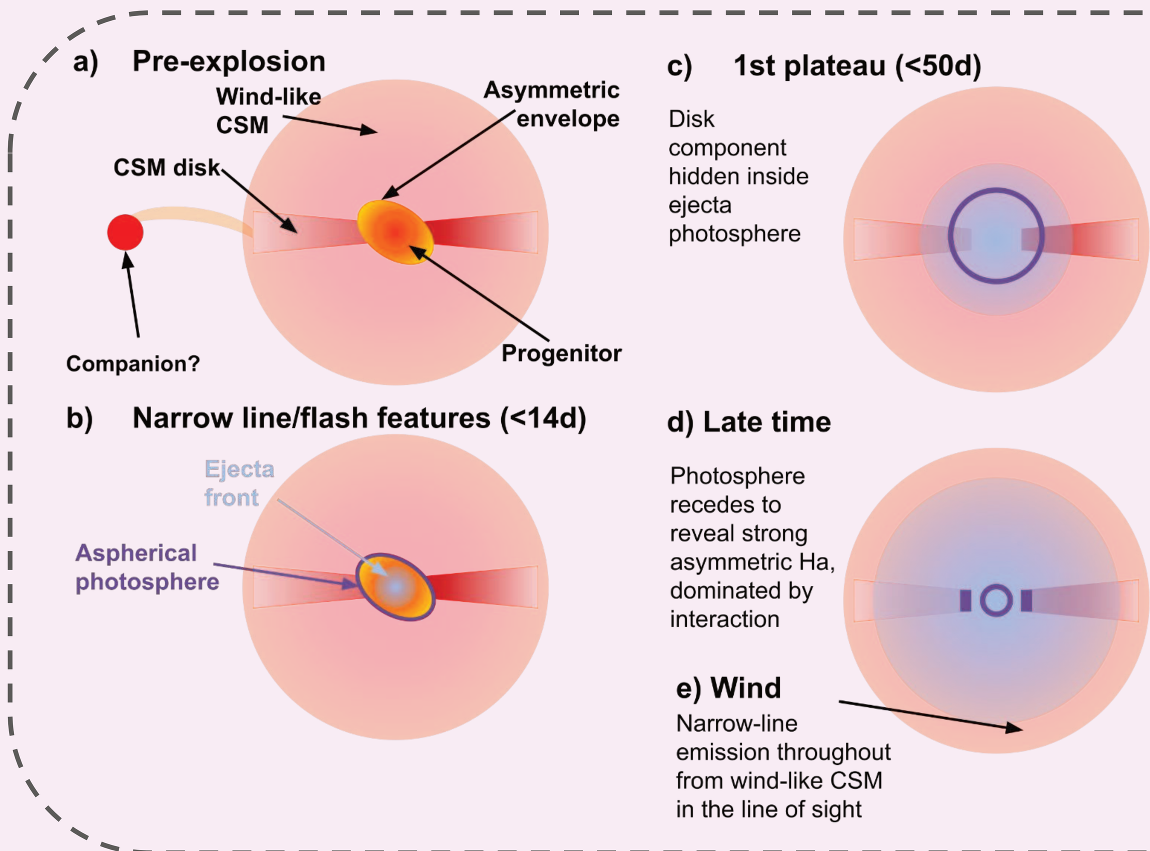


FIG 08 ● *Explosion scenario for SN 2024cld, from the early flash-ionisation phase, through to late-time interaction.*

SN 2024cld likely had a RSG/YHG progenitor.

06 Continued evolution

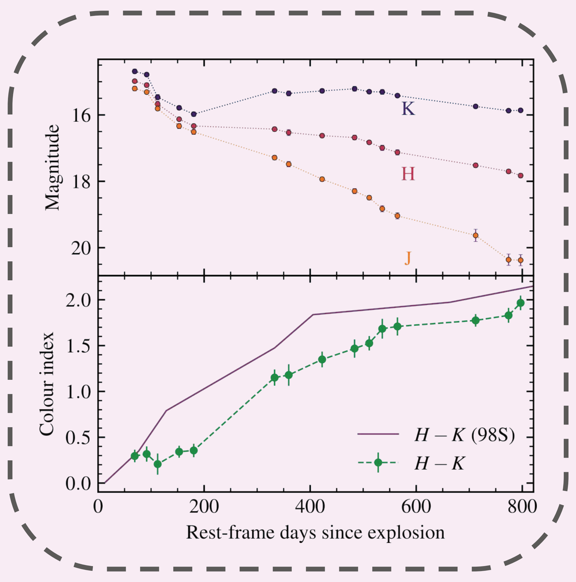


FIG 09 ● ***From +400d onwards, SN 2024cld has shown a clear K-band excess****, and dust-related features in a late time NIR spectrum. Awarded* ***X-shooter, ERIS, and JWST time*** *will constrain the dust properties and assess the dust yield of this exotic family of SNe.*

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